

Blindspot Report — Post-Round-2 Outputs

Outputs audited: (1) §33 measurement-invariance pipeline (code/36/37/38, `_invariance_results.tsv`); (2) the 2026-05-01 / 2026-05-02 composite-density, item-decomposition, and item-density figures for Minority Support, SD: Primary Out-group, and SD: General Out-group (scripts 39–50); (3) the §51 paired Estonian Primary-vs-General test (code/51, `viz/fig_sd_primary_vs_general_*.jpg`); (4) the group-organized comparison figures (scripts 54-55).

Stated narrative arc since Round 2: (A) Estonian community is fragmenting on multiple intergroup dimensions while Russian community is broadly liberalizing; (B) Estonians' attitudinal shift is selectively targeted at Russian-speakers, not at out-groups in general — supported by the §51 sign-reversal of Estonian Primary vs. General SD.

Date: 2026-05-02

The Blindspot protocol is a structured perceptual audit that asks whether the author can still see what is in front of them — features in the output that have become invisible due to familiarity. It complements but does not replace the Referee 2 audit (Round 3, 2026-04-29) which checked whether the code is correct. This Round 3 Blindspot focuses on outputs produced between 2026-04-30 and 2026-05-02 — measurement invariance plus the new wave of distribution-shape visualizations and the Estonian Primary-vs-General sign-reversal claim.

Overall Verdict

CONDITIONAL — three substantive findings stand on solid measurement footing, two depend on a measurement caveat that has been documented but not yet propagated into the figure narratives, and one item-level pattern is being underplayed in the current write-up. No data-coding errors, no falsified claims. The single most important action item is to flag SD: General's item-set change explicitly wherever the within-General d's are cited, because the 'Russian liberalization' story leans heavily on a within-General d that is partially confounded with construct change.

FLAG 1 — Sign-reversal claim: solid on the Primary side, leakier on the General side

The §51 paired test established a striking sign-reversal in Estonian respondents: in 2020 Estonians rated Russian-speakers as LESS socially distant than 'new immigrants' ($M_{Pri} = 2.51 < M_{Gen} = 2.91$, $d_z = -0.47$, ***); in 2023 the rank order flipped ($M_{Pri} = 2.92 > M_{Gen} = 2.70$, $d_z = +0.24$, ***). The within-year paired tests are clean, and the §54 visualization makes the reversal immediately visible. This is a strong finding.

The vulnerability is in how the reversal is being attributed. The current narrative treats both halves of the swing as substantive: Estonians became MORE distant from Russian-speakers AND MORE accepting of generic out-groups. The first half is well-supported — within-Estonian SD: Primary $d = +0.42$ *** is built on items that hold scalar measurement invariance longitudinally (per §33: $\Delta\chi^2(2) = 4.5$, $\Delta CFI = -0.002$, $\Delta RMSEA = +0.0075$ — well within Cheung & Rensvold thresholds). The second half — within-Estonian SD: General $d = -0.23$ — compares a 3-item 'new immigrants' battery in 2020 against a 6-item 'other Europeans + non-Europeans' battery in 2023. CLAUDE.md flags this as 'cross-year measures different construct.' This caveat is missing from the §51 / §54 / §55 writeups.

Recommended reframing: present the falsification claim as one-sided. 'Estonians selectively hardened toward Russian-speakers' is well-supported. 'Estonians simultaneously liberalized toward generic out-groups' rests on a non-comparable construct and should be qualified or demoted to a methods-chapter footnote. The cleanest version of the sign-reversal claim is: 'In 2020, the rank order Primary < General held (Estonians regarded Russian-speakers as more socially proximate than abstract immigrants). In 2023, the Primary distribution had moved past the General distribution. The Primary-side

movement is the substantive driver; the General-side movement may reflect both attitudinal shift and construct re-specification.'

FLAG 2 — Measurement-invariance caveats are documented but not propagated to figure narratives

§33 measurement-invariance testing established that scalar invariance HOLDS for SD: Primary Out-group longitudinally (both groups), but FAILS for Belief in Inevitable Conflict, Minority Support Inclusion, and most other composites under longitudinal and multi-group comparison (8 HOLDS, 2 PARTIAL, 18 FAILS across 28 comparisons). The substantive implication — that some portion of the observed group-level d's may reflect intercept variation rather than purely attitudinal differences — was filed in the methods documentation. It has not made its way into how the recent figures are being interpreted.

Specific cases where this matters:

(a) BiC within-Russian d = -0.32 * ('Russians decreased belief in inevitable conflict')**

Longitudinal scalar invariance FAILS for Russians ($\Delta\chi^2(3) = 28.7$, $p < .001$, $\Delta CFI = -0.046$). The 'Russians decreased conflict belief' claim — central to dissolving the §29 paradox — is robust to direction (the recoding was independently verified) but not strictly comparable as a latent-mean difference. Worth noting in the chapter.

(b) Minority Support Inclusion within-Estonian d = -0.15 ** and within-Russian d = +0.04 (ns)

Longitudinal scalar invariance FAILS for both groups. The 'Russians stayed flat / Estonians modestly declined' framing rests on score differences whose measurement-intercept comparability is not established. The item-level decomposition partially compensates (it shows specific items drove the Estonian decline), but composite-level interpretation should be qualified.

(c) SD: Primary Out-group within-Estonian d = +0.42 * ← exception worth highlighting**

Longitudinal scalar invariance HOLDS for both groups. This is the rare composite where a strong claim about latent attitude change can be made with confidence. The fact that the headline Round-3 finding (Estonians selectively hardened toward Russian-speakers) sits on the most measurement-robust composite in the project is a lucky alignment that should be cited as supporting evidence rather than left implicit.

FLAG 3 — Russian sample composition shift (609 → ~520) still unresolved

Round 2 (§29 FLAG 1) raised that the Russian respondent N dropped from 609 (2020) to ~520 (2023, varies slightly by composite due to DK patterns), and that this shift was undocumented. Eight days later, no robustness check has been performed. The 'Russians broadly liberalized' narrative — built on within-Russian d's of -0.32 (BiC), -0.56 (SD General), +0.04 (Minority Support stable), and -0.07 (SD Primary stable) — is potentially compositional rather than attitudinal: if more conservative Russian respondents selectively dropped out between waves, every within-Russian estimate is biased in the liberalizing direction.

The Round 2 Russian-edu-pathway robustness check (§31) addressed within-2023 sub-group consistency, but does not test whether 2020 and 2023 Russian respondents are demographically comparable. The minimum useful check is: (i) cross-tab age, education, and region for the Russian sub-sample in 2020 vs. 2023; (ii) if a noticeable shift exists, re-estimate the within-Russian d's with post-stratification weights or a covariate-adjustment regression. This has been pending since April 29 and should be closed before any of the within-Russian narratives are committed to the thesis chapter.

FLAG 4 — The Ukrainian dimension is structurally absent from the General out-group story

The 2020 → 2023 General Out-group shift — within-Russian $d = -0.56^{***}$, within-Estonian $d = -0.23^{***}$ — is being reported as 'both groups became more accepting of general out-groups.' Two structural facts complicate this narrative:

- The 2020 'new immigrants in last 5 years' item was administered in a period when Estonia had very few new immigrants. It measured an abstract category.
- The 2023 General battery measures 'other Europeans + non-Europeans' — and explicitly EXCLUDES the Ukrainian items per the 2026-03-23 decision (Russians held asymmetrically negative views of Ukrainian refugees, which masked otherwise positive general-out-group attitudes).

The exclusion was methodologically defensible. But it means the 'Russians liberalized toward general out-groups' claim is partial: Russians liberalized toward generic Europeans and non-Europeans, while remaining notably more negative than Estonians toward the most salient post-2022 immigrant population. A reader unfamiliar with the methods will mis-read the General SD decline as a generalized softening of attitudes — which the underlying data do not support. The Ukrainian item-level distributions are sitting in the data and could be plotted separately to make this asymmetry visible without contaminating the composite.

FLAG 5 — Item 3 'Understand opinions' shows an asymmetric within-group divergence that has not been pursued

From the Minority Support item-level decomposition (script 40, May 1):

Item	Within Estonian d	Within Russian d	Note
Q68_1 'Involve in economy'	-0.01 ns	+0.05 ns	stable both sides
Q68_2 'Involve in governance'	-0.18 ***	-0.08 ns	Estonians declined modestly
Q68_3 'Understand opinions'	-0.16 **	+0.23 ***	DIVERGENT — only opposite-sign item

Item 3 is the only Minority Support item where Estonian and Russian respondents moved in OPPOSITE directions, both significantly. The substantive interpretation is striking: 'understanding the opinions of other nationalities' grew more important to Russians while becoming less of a priority to Estonians. From each group's own perspective, this means Russians became more open to understanding Estonian-language society while Estonians became less open to understanding Russian-language society. This is exactly the asymmetric divergence pattern that the headline Estonian-vs-Russian narrative would predict — and yet it sits in a single item-level row of an appendix figure, not in the main results discussion.

Recommended action: promote Q68_3 to a stand-alone result. The item is conceptually closer to 'cultural curiosity' / 'cross-language receptiveness' than the other two governance/economy items, and the divergence is qualitatively different from the broader fragmentation pattern. Worth a paragraph in the discussion.

FLAG 6 — Within-Estonian SD: Primary movement is concentrated in PUBLIC contexts, not the most intimate one

From the SD: Primary item-level decomposition (script 43, May 1):

Item	Within Estonian d	Within Russian d
Neighbors	+0.41 ***	-0.04 ns
Work / study	+0.44 ***	-0.01 ns
Marriage in family	+0.22 ***	-0.11 (p=.08)

Estonians' increased social distance from Russian-speakers is half as large at the most intimate social-distance item (marriage in family) as at the public-civic items (neighbors, work/study). This asymmetry is consistent with two interpretations: (i) the shift is largely a public-life phenomenon — civic and workplace boundary-setting — that is less

reflected in personal/family decisions; (ii) the marriage item is at floor for many Estonian respondents in both years (modal category 1 = would not accept), so there is less room for a distance increase to manifest.

Both interpretations matter. (i) is theoretically interesting and worth pursuing as a sub-finding. (ii) is a measurement-floor caveat that could be checked against the item distribution. The current writeup presents the Estonian Primary-d as a single composite movement; surfacing the item-level gradient would sharpen the substantive interpretation.

FLAG 7 — The 'Estonian shift is not generalized' falsification logic is weaker than presented

The §29 reframing — and the §51 paired test that operationalized it — argues that the divergence between within-Estonian SD: Primary (+0.42 ***) and SD: General (−0.23 ***) constitutes a falsification test for the war-shock hypothesis: if Estonians had simply hardened against all out-groups, both within-d's should be positive. The fact that they go in opposite directions, the argument runs, shows that the hardening is selectively targeted.

The argument has two soft links:

- The General d also goes in opposite directions for Russians (−0.56) and Estonians (−0.23). If the General item-set change drove most of the General d for both groups (a strict construct-change interpretation), then the Estonian Primary increase remains the only substantive change being reported — the falsification test is reduced to 'Estonians moved on Primary, period.' That is a weaker claim than 'Estonians moved on Primary while not moving (or moving the other way) on General.'
- Even if the General-side d's are accepted as substantive, a 'liberalized toward generic Europeans + non-Europeans' shift while 'hardening toward the specific historically-salient out-group' is compatible with a war-shock-driven schema split (Russian-speakers tagged as threat, generic Europeans tagged as solidarity). It is not strictly disconfirming for war-shock, only for naïve generalized-hardening.

Recommended action: weaken the falsification claim from 'falsifies generalized hardening' to 'rules out non-targeted hardening,' which is what the design actually supports.

Confirmations — what the new outputs do well

- The §51 paired test was the missing piece behind the Round 2 §29 falsification framing. Adding a within-respondent test transforms the Round 2 'two within-group d's go opposite directions' into 'the rank order between the two out-group categories actually flipped within Estonians.' The Estonian sign-reversal is the single sharpest finding the project now has, and it now has a proper test behind it.
- The script-50 4×3 SD General item-density figure handles the awkward 2020/2023 item-set asymmetry cleanly — the 'one 2020 immigrant target splits into two 2023 sub-targets' framing is the most honest way to display a non-comparable battery, and the figure makes the asymmetry legible rather than hiding it.
- Distribution-shape figures (composite density + item-density across Minority, SD Primary, SD General) consistently surface the variance / dispersion patterns alongside the means. The Estonian dispersion-grows pattern that emerged in Round 2 (Superordinate Identity, Group ID, BiC) now also shows up in Minority Support (Levene's $p = .001$) and SD Primary composite ($p = .001$) and SD Primary Work / study item ($p = .003$). Five different dimensions of intergroup attitudes show the same fragmentation signature on the Estonian side. That's a strong convergent pattern.
- The exploratory in-group SD investigation (scripts 45-47) was correctly diagnosed as structurally non-interpretable ($N = 21-51$ per cell, almost certainly a survey-routing artifact) and dropped without further pursuit. Good ergonomics.

Action Checklist (in priority order)

- 1. Reframe §51 / §54 / §55 narrative: emphasize within-Estonian Primary increase as the substantive driver; qualify the within-General decrease with the construct-change caveat.

- 2. Cite the §33 longitudinal scalar-invariance HOLDS for SD: Primary as supporting evidence for the headline finding (turn the methodology lucky-alignment into a cited strength).
- 3. Add the §33 scalar-invariance FAILS caveat to the methods chapter for BiC and Minority Support within-group findings.
- 4. Run the long-pending 2020 vs. 2023 Russian sample composition robustness check (Round 2 FLAG 1, still open).
- 5. Surface Q68_3 'Understand opinions' divergence (Estonians -0.16^{**} , Russians $+0.23^{***}$) as a stand-alone discussion result — it is a different kind of asymmetry from the rest.
- 6. Explicitly address the Ukrainian-item exclusion in the General-out-group narrative: either plot the dropped Ukrainian items separately, or add a paragraph clarifying that 'Russian liberalization toward general out-groups' excludes the most policy-salient post-2022 group.
- 7. Weaken the falsification claim from 'falsifies generalized hardening' to 'rules out non-targeted hardening' — preserve the substance, sharpen the epistemic standing.
- 8. (Smaller) Note the SD: Primary public-vs-intimate item gradient (Estonian Marriage $d = +0.22$ vs. Neighbors / Work $+0.41 / +0.44$) — interesting sub-finding worth one or two sentences.